

Once the source begins to evolve, Pressure can still flip the sign independently

R0.70K narrowed the next step to one precise question: can the evolution of the genuine low-frequency strain source compensate for the positive variance of the normalized vorticity shape? This version derives the complete coupled ledger. The local quadratic term already takes both signs; a further support-separation theorem and an exact periodic sign pair show that no nontrivial instantaneous compensator depending only on the current local source and local shape can have a universal sign.

Status · R0.70L exact derivation, independent internal review PASS, and formal figure completed

Instantaneous local-compensator branch closed

Version v0.70L · 2026-08-24
Exact generators: all 4 groups passed
Figure validation: all 20 checks passed
Exact periodic witnesses: same local data, opposite derivatives
Public characterization: structural initial-surface no-go result

00 / EXACT CONCLUSION

Evolving the genuine source does not close the sign barrier

Along trajectories of the filtered velocity U , take the true strain source

$$\Sigma(t) = S(U)(X(t), t), \quad \dot{X} = U(X, t),$$

and continue using the trace-normalized shape from R0.70K

$$R = \frac{Q}{\text{tr } Q}, \quad B = R - \frac{1}{3}I, \quad q = \Sigma : B.$$

The full derivation retains pressure, viscosity, the subgrid Hessian, and the covariance residual. The result is not a new dissipation law: the local quadratic term itself is sign-indefinite, and while keeping the local (Σ, B, q) fixed, pressure can arbitrarily change $\dot{q}$.

Route decision. The branch “construct a universal Lyapunov scalar from the current local strain and current local vorticity shape” is closed. Further work must add nonlocal pressure, spatial integration, history, or states from adjacent scales.

01 / RESOLVED SOURCE EQUATION

Every term in the filtered-strain source must be retained

If $\tau = \overline{u \otimes u} - U \otimes U$, then along trajectories the exact trace-free source equation is

$$\dot{\Sigma} = -(\Sigma^2)^\circ - \frac{1}{4}(\Omega_* \otimes \Omega_*)^\circ - H_*^\circ + \nu(\Delta S)_* - K_{\tau,*}^\circ.$$

Here $H = \nabla^2 P$, while

$$(K_\tau)_{ij} = \frac{1}{2} \partial_a (\partial_j \tau_{ia} + \partial_i \tau_{ja}).$$

The allocation between the pressure Hessian and the subgrid Hessian changes with the filtering convention; only their complete combination has a consistent meaning in the equation. Accordingly, this page does not replace any term with a local eddy-viscosity model.

02 / COUPLED CORRELATION LEDGER

The coupled evolution writes the gap as six checkable contractions

$$\begin{aligned} \dot{q} = & B : \Sigma^2 + \frac{2}{3} |\Sigma|_F^2 - 2q^2 - \frac{1}{4} B : (\Omega_* \otimes \Omega_*) - B : H_*^\circ \\ & + \nu B : (\Delta S)_* - B : K_{\tau,*}^\circ + \Sigma : \mathcal{T}_B(F_{\text{err}}). \end{aligned}$$

Denote the first local combination by $\mathcal{Q}(\Sigma, R)$. For the same strain $\Sigma = \text{diag}(2, -1, -1)$, choosing $R = e_1 \otimes e_1$ gives $\mathcal{Q} = -2$, while choosing $R = e_2 \otimes e_2$ gives $\mathcal{Q} = 1$. Thus, even before pressure enters, the purely local quadratic term has no definite sign.

Information	Resolved at this stage?
Exact coefficients and signs	Independently recomputed
Definite sign for the local quadratic term	No; exact positive and negative witnesses exist
Pointwise sign of pressure	No; its sign can be flipped independently
Nonlocal or historical compensation	Still open

Every local source direction is detectable by pressure duality

The support-separation construction of Ro.70J uses smooth velocity packets outside a buffered core to vary the symmetric trace-free pressure Hessian at the core arbitrarily, while leaving the core source Σ , normalized covariance B , cutoff motion, and pressure-free evolution of B unchanged.

Therefore, for any C^1 instantaneous scalar $\Phi(\Sigma, B)$, the pressure contribution is

$$\dot{\Phi}|_H = -D_\Sigma \Phi : H_*^\circ.$$

If it is required to have the same one-sided sign for all smooth initial data within the stated scope, then necessarily

$$D_\Sigma \Phi = 0.$$

This does not mean that “pressure cannot be studied.” It means only that if a compensator has no access to nonlocal pressure yet depends nontrivially on the local source, pressure alone can destroy its universal sign.

The same local state yields exact derivatives of opposite signs

The report constructs two smooth periodic divergence-free initial data u_- and u_+ . At the origin and initial time, they have the same

$$\Sigma = \text{diag}(1, -1, 0), \quad R = \text{diag}(1/2, 0, 1/2), \quad B = \text{diag}(1/6, -1/3, 1/6), \quad q = 1/2.$$

The local gradient, source viscosity, and normalized-covariance evolution are also identical. Only the pressure contraction changes; exact Fourier inversion gives

$$\dot{q}_- = \frac{3901}{2040} > 0, \quad \dot{q}_+ = -\frac{1283}{2040} < 0.$$

The two initial data have equal energy, and each generates a genuine short-time smooth Navier–Stokes solution. The witness occurs on the initial surface; it does not claim that the long-time trajectories remain matched.

The no-go theorem leaves five larger state spaces

What remains unexcluded is pressure orthogonality after spatial integration and its localization commutators, an explicit nonlocal pressure variable, a coupled pressure–subgrid ledger across adjacent filter scales, Lagrangian history, and the matrix pullback by the strain propagator.

The smallest next threshold among them is

$$\dot{G} = \Sigma G, \quad \hat{Q} = G^{-1} Q G^{-\top}, \quad \dot{\hat{Q}} = G^{-1} F_{\text{err}} G^{-\top}.$$

At the matrix level, this pullback exactly eliminates the constant source, but it introduces a history-dependent condition number. Ro.70L retains it only as an Ro.70M candidate and does not present the condition-number estimate as a closed conclusion.

Pressure is the only switch in the matched source–shape sign pair

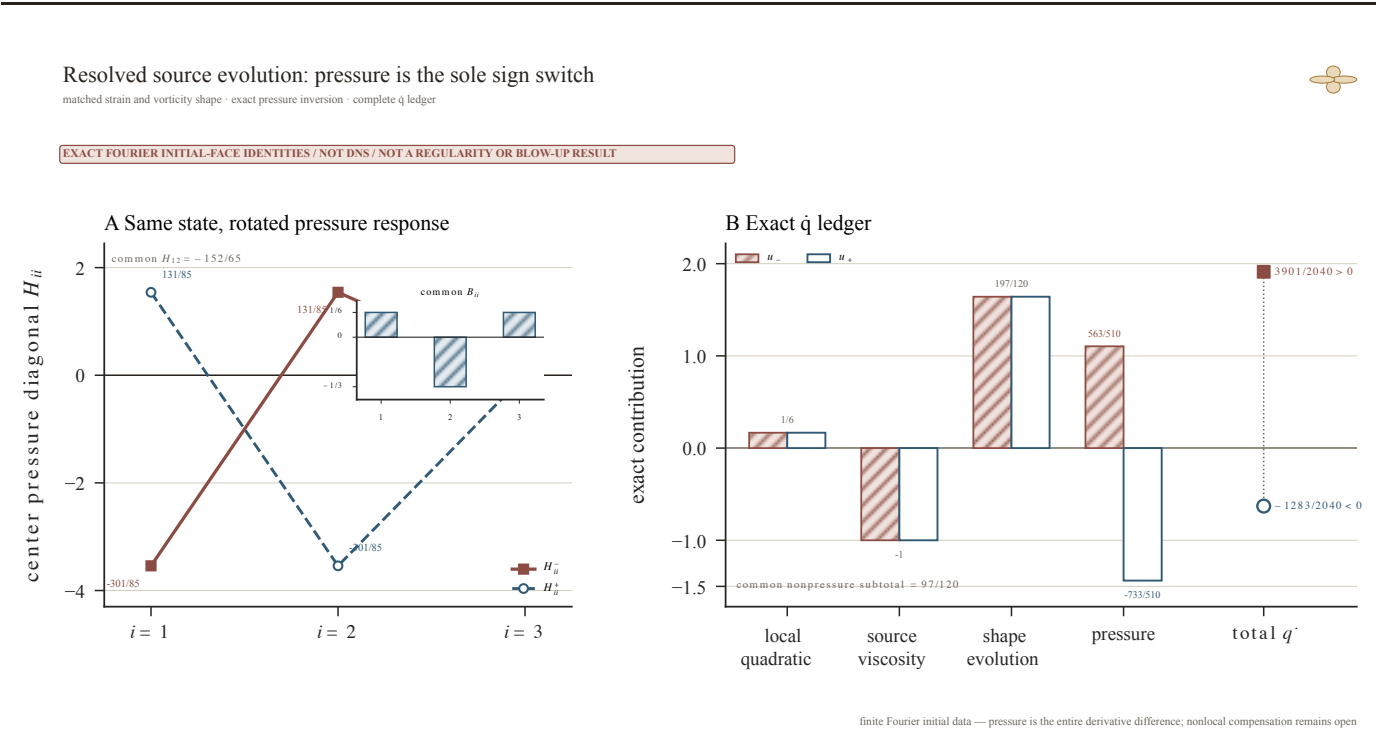


Figure Ro.70L. A: The two smooth periodic initial data have the same Σ and B , while the two diagonal axes of the pressure Hessian are interchanged. B: The local-gradient, source-viscosity, and covariance-shape contributions agree term by term; only the pressure term changes sign, yielding opposite $\dot{q}$. All data are exact finite-Fourier identities on the initial surface, not DNS.

The value lies in exactly ruling out a natural but undersized state space

The significance of Ro.70L is not that it yields a regularity estimate, but that it avoids further enumeration of local polynomials in (Σ, B) . The nonlocality of pressure is classical; this stage adds a no-go theorem for local functionals under the stated assumptions, together with a periodic Navier–Stokes witness having exactly matched local data and exactly opposite derivatives.

A bounded audit of eleven primary sources covers the exact filtered velocity-gradient equation, conditional pressure closure, the Germano identity, Gaussian scale decomposition, and strong-vorticity statistics. None provides the deterministic local compensating sign required here. This is a conclusion from a finite search, not a proof of nonexistence across the entire literature.

The next step will not search for more local scalars. I will only examine the full residual pullback by the strain propagator, and whether it can return to physical variables without presupposing control of the accumulated critical strain.

This is a structural no-go result on the initial time slice, not a regularity theorem

PROVED Complete source and correlation ledger Pressure, viscosity, vorticity, and subgrid terms are all retained.	PROVED Pressure threshold for local functionals Valid under the buffered-core and stated filtering assumptions.
OPEN Nonlocal and historical compensation Pressure variables, spatial integration, and adjacent scales remain open.	OPEN From energy to critical control The weak-solution limit, denominator mass, and fixed positive time remain unresolved.

The theorem does not automatically cover noncompact Gaussian or strict Littlewood–Paley filters, spatially averaged sources, pressure-dependent truncated acceleration, functionals involving pressure or history, nor does it claim an optimal result on a fixed energy ball.

This version constructs no singularity, proves no global smoothness, and is not a partial solution of the Millennium Problem.

The report, independent audit, literature audit, certificates, and formal figure package can be checked separately

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The generator certifies only finite Fourier, matrix, and rational-arithmetic calculations. The continuum filtering derivation, support-separation proof, short-time smooth solutions, and literature interpretation are separate analytic arguments and are not replaced by the computer certificate.